INSURANCE CLAIMS INTELLIGENCE

Project Documentation & Future Reference Guide

Project type	Portfolio project | Insurance / Financial Services Analytics
Primary role	Power BI Data Analyst / BI Analyst
Core tools	Excel, Python, Statistics, Power BI, DAX
Dataset	Synthetic practice dataset; 6 business tables, approximately 283K rows in total
Dashboard	Three pages: Executive Overview, Fraud & Risk Analysis, Financial & Claims Analysis

Why this guide exists

This document records the business problem, dataset, data model, measures, dashboard pages, design decisions, known limitations and next steps. Read it after a break of several months to quickly remember what the project does, how it is built and what still needs to be validated.

Project status note: The synthetic Excel dataset has been prepared and the three Power BI page designs are visible in the screenshots shared in the conversation. Some metric labels and visuals still need the corrections listed in this guide. Python cleaning/statistical analysis and final validation should only be marked complete after you actually run and document them.

Prepared as a practical handover note for your future self

1. Project at a glance

The Insurance Claims Intelligence project is an end-to-end analytics portfolio project. It uses insurance policy, customer, claim, payment and investigation data to help an insurer monitor claim volumes, claim costs, possible fraud-risk indicators and payment performance.

The project is designed to demonstrate a BI analyst workflow: understand a business problem, inspect and prepare data, define reliable metrics, build a relational data model, create an interactive Power BI report and explain business insights in plain language.

Business scenario

An insurance company receives claims across different claim types and regions. Management needs to understand how many claims are being submitted, how much money is being claimed and paid, which claim categories contribute most to costs, and which claims have been flagged for further review. The report provides a monitoring view; it is not an automated fraud verdict or a claim approval system.

Main project objectives

- Measure claim volume and claimed financial exposure.
- Monitor claims marked with a potential-risk / fraud flag.
- Compare claim patterns by type, status, region and previous-claim history.
- Understand payment amounts and payment statuses.
- Summarise investigation workload and outcomes.
- Present the results in a clear, interactive three-page dashboard.

2. Business questions this project answers

Area	Business question
Claims overview	How many distinct claims have been recorded? What is their total claimed amount? What is the average claim amount? How do claim counts change month by month?
Claim categories	Which claim types have the highest number of claims? Which claim types contribute the greatest total claim amount? Which types have the most flagged claims?
Claim status	How many claims are Approved, Closed, Submitted, Under Review or Rejected? What is the claimed amount associated with each status?
Potential fraud / risk	How many claims are flagged? What percentage of claims are flagged? How do flagged claims vary by claim type, region and previous-claims count?
Investigations	How many distinct claims have an investigation record? What is the distribution of investigation statuses and outcomes?
Regional patterns	Which business regions have the highest claim frequency and flagged-claim counts?
Payments	How much has been recorded in payment transactions? How many payment records exist? What share of payment records are Paid, Pending or Failed?
Business prioritisation	Which claim categories, regions or patterns may deserve closer operational review? Which areas contribute most to claim volume or financial exposure?

Important interpretation: the dashboard helps identify patterns and exceptions for human review. A flagged record is not proof of fraud. The synthetic flag is a practice field, not a validated real-world fraud label.

3. Dataset and workbook structure

The practice workbook is named **Insurance_Claims_Intelligence_Raw.xlsx**. It contains a README sheet and six business data sheets. The dataset is synthetic and was generated for learning and portfolio demonstration, not taken from a real insurer.

Sheet	Approx. rows	Key	What it contains
Customers	30,000	CustomerID	CustomerAge, Gender, Region, CityTier, CustomerSince, AnnualIncomeINR
Policies	40,000	PolicyID	CustomerID, PolicyType, PolicyStartDate, PolicyEndDate, AnnualPremiumINR, CoverageAmountINR, PolicyStatus
Claims	100,000	ClaimID	PolicyID, ClaimTypeID, IncidentDate, ReportedDate, ClaimAmountINR, ClaimStatus, PreviousClaimsCount, FraudFlag
Payments	98,117 (approx.)	PaymentID	ClaimID, PaymentDate, PaymentAmountINR, PaymentStatus
Investigations	15,000	InvestigationID	ClaimID, InvestigationDate, InvestigationStatus, InvestigationOutcome
Claim_Types	15	ClaimTypeID	ClaimTypeName, BusinessLine

How the tables fit together

Customers own or are associated with policies. Each policy can have multiple claims. Each claim has a claim type. A claim may have one or more payment records and may have one or more investigation records. This is why Payments and Investigations are child/detail tables of Claims.

Suggested Power BI relationships

Parent key	Relationship	Child key
Customers[CustomerID]	1 → many	Policies[CustomerID]
Policies[PolicyID]	1 → many	Claims[PolicyID]
Claim_Types[ClaimTypeID]	1 → many	Claims[ClaimTypeID]
Claims[ClaimID]	1 → many	Payments[ClaimID]
Claims[ClaimID]	1 → many	Investigations[ClaimID]

Use single-direction filtering from the parent table to the child table unless there is a clear reason to do otherwise. Confirm that each parent key is unique and that data types match before creating relationships.

Critical aggregation warning: do not merge Claims directly with Payments and Investigations and then sum Claims[ClaimAmountINR]. Multiple child rows can repeat the same claim and inflate the total. Keep the tables related in the model and calculate claim amounts from the Claims table.

4. Core Power BI measures (DAX)

These measures describe the intended calculations. Create them in Power BI using Modeling → New measure. Check table and column names against your imported model before pasting.

Total Claims

```
DISTINCTCOUNT(Claims[ClaimID])
```

Counts unique claims, not rows from a joined table.

Total Claim Amount

```
SUM(Claims[ClaimAmountINR])
```

Sum of amounts claimed, not necessarily amounts paid.

Flagged Claims

```
CALCULATE(DISTINCTCOUNT(Claims[ClaimID]), Claims[FraudFlag] = 1)
```

Counts unique claims marked with the simulated flag.

Average Claim Amount

```
DIVIDE([Total Claim Amount], [Total Claims], 0)
```

Average claimed amount per distinct claim.

Flag Rate

```
DIVIDE([Flagged Claims], [Total Claims], 0)
```

Format this measure as Percentage.

Investigated Claims

```
DISTINCTCOUNT(Investigations[ClaimID])
```

Distinct claims with at least one investigation record.

Total Payment Amount

```
SUM(Payments[PaymentAmountINR])
```

Sum of payment transaction amounts.

Payment Records

```
DISTINCTCOUNT(Payments[PaymentID])
```

Counts unique payment transactions.

Metric definitions to remember

Claim amount: the amount requested/recorded on a claim. **Payment amount:** a transaction amount recorded in the Payments table. These are different business concepts and should not be used interchangeably.

Flag rate: flagged distinct claims divided by all distinct claims in the current filter context. It is not the percentage of confirmed fraudulent claims.

Investigated claims: distinct ClaimID values appearing in Investigations. The number of investigation records can be larger because one claim may have multiple investigation rows.

5. Dashboard pages and what each visual communicates

Page 1 — Executive Overview

Purpose: give a manager a fast summary of overall claim activity and financial exposure.

Visual	Purpose / correct interpretation
KPI cards	Total Claims; Total Claim Amount; Flagged Claims; Average Claim Amount (or another correctly defined KPI).
Monthly Claims Trend	Shows how the number of claims changes by incident month. Use a proper date field and chronological month sorting.
Claims by Type	Compares claim volume across categories. If the chart axis says Total Claims, use a claim-count measure, not claim amount.
Claims by Status	Shows the number of claims in each status category.
Flagged Claims by Type	Shows which claim categories contain the largest number of flagged claims.
Slicers	Incident Date, Region, Claim Type and Claim Status, if added to the page.

Page 2 — Fraud & Risk Analysis

Purpose: help a reviewer explore the synthetic flagged-claim patterns and investigation workload. Avoid presenting the flag as a final fraud decision.

Visual	Purpose / correct interpretation
Flagged Claims	Distinct count of claims where FraudFlag = 1.
Flag Rate	Flagged Claims / Total Claims; display as a percentage.
Investigated Claims	Distinct claims with investigation records.
Flagged Claims by Type	Compares flagged claim counts across claim types.
Previous Claims Count	Shows flagged-claim distribution by PreviousClaimsCount, if the visual is filtered to flagged claims. Otherwise title it clearly as all claims.
Investigation Status	Distribution of investigation records by status; label it as record count if it counts rows.
Regional analysis	Use a bar chart for North, South, East, West and Central. A map is not reliable with these broad labels unless proper state/city coordinates are added.

Page 3 — Financial & Claims Analysis

Purpose: examine claim exposure and recorded payment activity.

Visual	Purpose / correct interpretation
Total Claim Amount	Sum of claim amounts from Claims.
Payment Amount	Sum of payment transaction amounts from Payments.
Average Claim Amount	Average amount per distinct claim.
Payment Records	Distinct payment transaction count.
Monthly Claim Amount	Claim amount by incident month; make sure months are sorted chronologically.
Claim Amount by Type	Treemap or bar chart showing total claim amount by category.
Payment Status	Payment records or payment amount by Paid, Pending and Failed; title should identify which one.

Visual	Purpose / correct interpretation
Claim Amount by Status	Use a clustered bar/column chart for independent status categories. Avoid a Waterfall chart unless values truly represent sequential additions/subtractions.

6. Visual design system

The current report uses a dark navy theme with blue and cyan accents. Keep the same visual language across all three pages so the report feels like one product.

Use	Hex	Application
Canvas background	#0B1220	Main report page background
Visual/card background	#151F32	KPI cards and chart panels
Blue	#2563EB	Primary chart highlight
Cyan	#22D3EE	KPI values / emphasis
Green	#22C55E	Positive or paid status, where appropriate
Red	#EF4444	Risk/failed/rejected status, where appropriate
Main text	#FFFFFF	Titles and key labels
Secondary text	#A7B4C8	Subtitles and axis labels

Layout guidelines

- Use a consistent 16:9 report canvas and the same page margins.
- Keep KPI cards aligned at the top; use the same height, title style and number formatting.
- Use descriptive chart titles that state the measure, for example “Flagged Claims by Region”.
- Avoid scrollbars inside charts where possible; show Top N categories or increase visual space.
- Use consistent font sizes, number units and decimal places.
- Add slicers in a consistent location and test whether they filter the visuals as intended.
- Use tooltips for extra context rather than overcrowding every chart.

7. Corrections identified from the screenshots

Item	Recommended correction
Executive Overview KPI	The first card showed “$100K” for Total Claims. If this is a claim count, remove the currency symbol. The fourth card appeared to repeat “TOTAL CLAIMS” while showing 44.25K; if it is average claim amount, rename it “AVERAGE CLAIM AMOUNT”.
Monthly Claims Trend	The screenshot axis used $K while the visual title says claim trend. Confirm whether it plots claim count or claim amount. Use count formatting for claim counts and INR/currency formatting for claim amounts.
Risk by Region map	The map displayed points across continents. The source dataset only has broad Indian region labels (North, South, East, West, Central), so the map is incorrectly geocoding them. Replace it with a regional bar chart, or add validated state/city-level geography before using a map.
Previous Claims Count	Make the chart title explicit. If it shows flagged claims, filter FraudFlag = 1 and title it “Flagged Claims by Previous Claims Count”.
Claim Amount by Status	The screenshot showed a Waterfall chart. Status categories are independent groups, not sequential changes; replace with a clustered bar or column chart.
Investigation Status	Confirm whether the donut counts investigation rows or distinct claims. If it counts rows, label it “Investigation Records by Status”.
Currency labels	Use INR/■ consistently for monetary values. Do not put a currency symbol on claim counts or record counts.

8. Data quality, assumptions and limitations

- The dataset is synthetic practice data, not real customer or insurer data.
- FraudFlag is a simulated binary indicator. Use “flagged”, “potential risk” or “requires review”; do not say “confirmed fraud”.
- There is no verified settlement date in the described Claims schema. Do not show Average Settlement Days unless a valid settlement date is added and the calculation is defined.
- There is no model-generated Risk Score in the described dataset. Do not show AI Risk Score or Average Risk Score unless you actually build, validate and document such a model.
- There is no detection-coverage field defined. Avoid a Detection Coverage KPI unless its numerator and denominator are explicitly defined.
- Payments and Investigations can contain multiple records per claim. Avoid fanout duplication when calculating claim-level metrics.
- Region is a broad category, not a geocoded location. Do not infer exact map locations from it.
- Synthetic values and relationships should be checked for nulls, duplicates, invalid dates, unexpected categories and orphan foreign keys before publishing.

9. End-to-end workflow: what the project involves

1. Business understanding

Define the insurer scenario, intended audience and business questions. Decide what “claim”, “flagged claim”, “claim amount” and “payment amount” mean.

2. Data understanding

Inspect workbook sheets, row counts, columns, keys, data types, missing values and relationships.

3. Python data preparation

Use pandas to load the workbook, inspect data, handle missing/duplicate/invalid records according to documented rules, standardise dates and categories, and export clean tables if required.

4. Statistical analysis

Calculate distributions and descriptive statistics such as count, mean, median, percentiles and spread. Compare claim amounts across types/statuses and explore previous-claim patterns. Avoid claiming causation from descriptive comparisons.

5. Data modelling

Load tables into Power BI and create one-to-many relationships using the correct primary and foreign keys. Keep claim-level and transaction-level measures at the correct grain.

6. DAX measures

Create reusable measures for total claims, claim amount, flagged claims, flag rate, average claim amount, investigated claims, payment amount and payment records.

7. Dashboard design

Build the three report pages with consistent theme, KPIs, charts, slicers, tooltips and navigation.

8. Validation

Reconcile Power BI totals to Excel/Python totals, test slicers, check month sorting, verify currency units and test relationships.

9. Insight writing

Write a short observation, evidence/metric, business implication and recommended next action for each meaningful finding.

10. Portfolio presentation

Explain the business problem, data, modelling, measures, design choices, limitations and insights. Clearly disclose that the dataset is synthetic.

10. Suggested interview explanation

“I created an Insurance Claims Intelligence dashboard using a synthetic insurance dataset containing customers, policies, claims, payments and investigations. The goal was to help an insurance business monitor claim volumes and financial exposure, analyse potential-risk flags, understand claim categories and regional patterns, and review payment and investigation activity. I modelled the data in Power BI, created DAX measures for distinct claims, claim amount, flag rate and payment metrics, and designed three report pages: Executive Overview, Fraud & Risk Analysis, and Financial & Claims Analysis. I also considered data-grain issues, especially avoiding duplicate claim amounts when payment or investigation records have multiple rows per claim. The fraud indicator is simulated, so I describe it as a flag for review rather than confirmed fraud.”

11. What to do next (completion checklist)

1. Correct the KPI titles and number/currency formatting on Page 1.
2. Replace the misleading regional map with a Flagged Claims by Region bar chart.
3. Replace the Waterfall chart for Claim Amount by Status with a clustered bar/column chart.
4. Check that Previous Claims Count is filtered and titled correctly.
5. Confirm Investigation Status counts records or distinct claims and label it accordingly.
6. Create/verify the Power BI relationships and all DAX measures.
7. Complete Python data profiling and cleaning; keep a short log of every cleaning decision.
8. Complete statistical analysis and record key findings with supporting numbers.
9. Reconcile totals across Excel, Python and Power BI.
10. Test slicers, interactions, sorting, tooltips and page navigation.
11. Add a short Insights / Recommendations section or text annotations to the report.
12. Prepare a portfolio README describing the synthetic data, tools, business questions and limitations.

12. Glossary

Term	Meaning
Claim	A request for compensation submitted under an insurance policy.
Claim amount	The amount recorded/requested for a claim in the Claims table.
Payment	A payment transaction recorded against a claim; one claim may have multiple payment records.
Flagged claim	A claim marked by the dataset's FraudFlag field for potential-risk review; not proof of fraud.
Flag rate	Flagged distinct claims divided by all distinct claims in the selected filter context.
Investigation	A review record associated with a claim; one claim may have multiple investigation records.
Primary key	A field that uniquely identifies a row in its table.
Foreign key	A field that links a row to a related row in another table.
DAX	Data Analysis Expressions; the formula language used for Power BI measures and calculations.
Grain	What one row represents, such as one claim or one payment transaction.

Final reminder

When you return to this project, first read Sections 2–5 to remember the business questions, data tables and measures. Then use Sections 7–11 to finish corrections and validation. Do not present any metric as completed or any flag as confirmed fraud unless the data and calculation genuinely support that statement.